Prototropic reactions of 3-phenyl and 3-hetaryl chromone derivatives in the ground and excited states.

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INTRODUCTION

3-phenylchromones (isoflavones) are widespread nalural substances. The biological activity of the isoflavonoids is high and diverse. This fact determines their usage in medicine as phytoestrogens, phytoalexines, etc.

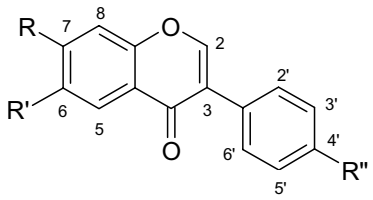
The hetaryl-analogs of isoflavonoids, which have been synthesized recently, also possess interesting biological activity. However, using these substances in pharmaceutical industry is impossible without sensitive spectral methods of their analysis, and, hence, without investigations of their fluorescent properties.

In our work the prototropic reactions and the spectral properties in the ground and in the excited states of several hydroxyisoflavones have been investigated.

pK_a and pK_b values have been esteblished spectrophotometically by acid-base titration of chromones in ethanol-water solutions.

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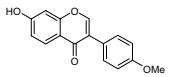
General formula



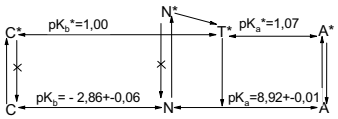
№ п/п	R	R‘	R“
1	-OH		$-OCH_3$
2	-OH	$-C_2H_5$	
3	$-OCH_3$	$-C_2H_5$	

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7-hydroxy-(4’-methoxy-3phenyl) chromone (Formononetin)



Scheme of the protolitic transformations

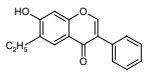


Spectral data of formononetin

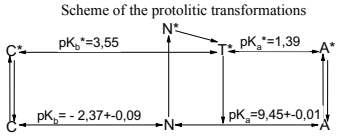
	λ_{abs}, nm/ ν, cm^{-1}	ε, $l \cdot mol^{-1} \cdot cm^{-1}$	λ_{fluor}, nm/ ν, cm^{-1}	$\Delta\nu_{st}$, cm^{-1}	φ, %	τ, ns
A	340,8/29340	12020	475,3/21040	8300	17,5	3,37
C	321,5/31100	11430	-	-	-	-
N, T*	302,7/33040	11110	470,0/21300	11740	1,2	1,57

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6-ethyl-7-hydroxy-3-phenyl chromone



Scheme of the protolitic transformations

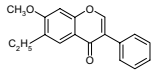


Spectral data of 6-ethyl-7-hydroxy-3-phenyl chromone

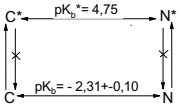
	λ_{abs}, nm/ ν, cm^{-1}	ε, $l \cdot mol^{-1} \cdot cm^{-1}$	λ_{fluor}, nm/ ν, cm^{-1}	$\Delta\nu_{st}$, cm^{-1}	φ, %	τ, ns
A	348,2/28720	15220	491,6/20340	8380	12,1	2,19
C	339,7/29460	19340	453,3/22030	7430	15,3	3,76
N, T*	308,0/32460	11750	519,8/19240	13220	1,73	0,71

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6-ethyl-7-methoxy-3-phenyl chromone



Scheme of the protolitic transformations



Spectral data of 6-ethyl-7-hydroxy-3-phenyl chromone

	λ_{abs}, nm/ ν, cm^{-1}	ε, $l \cdot mol^{-1} \cdot cm^{-1}$	λ_{fluor}, nm/ ν, cm^{-1}	$\Delta\nu_{st}$, cm^{-1}	φ, %	τ, ns
N	308,0/32460	14280	-	-	-	-
C	345,1/28980	19400	439,4/22760	6220	11,4	3,26

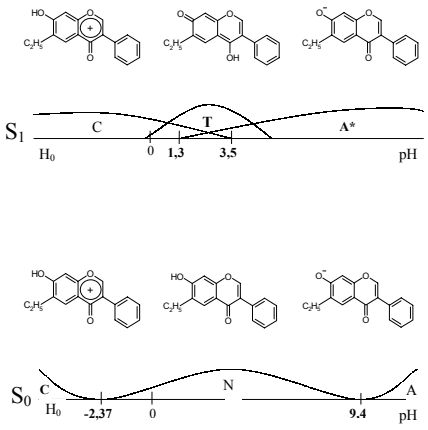
pK* in the excited state have been evaluated by Forster eqn:

$$pK_a^* = pK - \frac{hc}{kT}\left(\tilde{\nu}_{HA} - \tilde{\nu}_{A^-}\right),$$

where h – Planck constant,
c – speed of light,
k – Bolzmann constant,
T – absolute temperature.

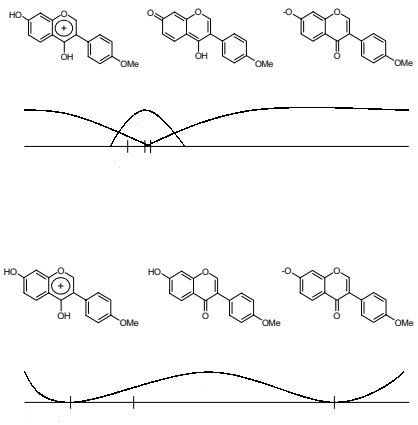
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Scheme of the 6-ethyl-7-hydroxy-3-phenyl chromone’s prototropic forms, depending on pH/H_0, in the ground and in the excited states



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Scheme of the 7-hydroxy-(4’-methoxy-3phenyl) chromone’s prototropic forms, depending on pH/H_0, in the ground and in the excited states



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Conclusions

1. Analysis of absorption and fluorescence spectra in acidity interval from H_0 ~ - 9 to pH ~ 12, as well as in solvents of different proton-donating ability, showed that prototropic reactions of chromones in the excited state differ substantially from the ones in the ground state. In the case of 7-hydroxychromones fluorescence of anionic (***A***), cationic (***C***) and phototautomer (***T***) forms was observed.

2. Emission of neutral form was not found. This phenomenon may be explained by the presence of forbidden n-π electronic transition (~370-380 nm) possessing lower energy than the first π-π transition (305-310 nm).

3. Acid-base titration of chromones in ethanol-water solutions and the following estimations of their pK_a and pK_b in the ground and in the excited states have shown that excitation leads to the increase of carbonyl group basicity and to the substantial increase of 7-hydroxygroup acidity (up to 8 orders of magnitude). Consequently, presence of anion and tautomer fluorescence in large interval of pH may be explained by photodissociation process. Decrease of proton-donating solvent polarity leads to fluorescence quenching of anion, but does not have influence on phototautomer emission. Consequently, photodissociation in non-polar media results, most probably, in quasi-intramolecular proton transfer and in formation of phototautomer, which is more stable in non-polar media in comparisin to that of anion.